4. MODEL OPTIMIZATION AND TUNING PHASE

4.1 Hyperparameter Tuning

Parameters tuned:

- max_depth
- min_samples_split
- min_samples_leaf

Technique:
GridSearchCV

4.2 Performance Improvement

Optimized Decision Tree achieved:

- Improved recall
- Reduced overfitting

4.3 Final Model Selection Justification

Decision Tree selected because:

- Best fraud recall
- Fast inference time
- Simple architecture
- Business interpretability

Deployment Summary

Model Saved Using

```
pickle.dump(dtc, open("dtc.pkl", "wb"))
```

Web Deployment

- Streamlit UI
- ngrok tunneling (Colab testing)

□ Advantages

- Handles large-scale data
- High fraud recall
- End-to-end ML lifecycle
- Real-time prediction ready

□ Limitations

- No SMOTE used
- No probability threshold tuning
- No XGBoost tested
- No SHAP explainability

□ Future Improvements

- Add XGBoost model
- Implement SMOTE
- Optimize threshold via PR curve
- Deploy on Streamlit Cloud
- Add model explainability (SHAP)
- Convert to Flask REST API